IDS 702 Project Proposal

Team member names here

Project Proposal/Statistical Analysis Plan

Link to your team's Github Repo: https://github.com/chen04301121-cmyk/ids702-ai-unemployment

Dataset Option 1

Project Title:

Data citation (use APA formatting or something similar):

Original data collection date (when was the dataset published?):

Data pull date (when did you download the data you will use?): IMF Labor Statistics (LS): November 3, 2026; IMF AI Adoption Index (AIAX) and World Economic Outlook (WEO): November 4, 2026.

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1 Research Question

Write the research question you are interested in investigating in your analysis. Consult this helpful guide for writing a good research question. Go beyond "is there a relationship between X and Y" to incorporate some statistical complexity, e.g., "is there a relationship between X and Y and does this relationship change over time?" You may also want to provide a second RQ as a backup and I can provide feedback on both options. Remember that you are required to use an inferential research question.

2 Study Population

Describe the observation units and how the data were collected.

3 Outcomes, Exposures, and Additional Variables of Interest

Add rows as necessary. Use the data dictionary for the coding specifications, e.g., 1=Diagnosed with CHD, 0=No diagnosis

3.1 Primary Outcome(s)

Variable Name	Description	Coding specification

3.2 Primary Independent Variable(s)

Variable Name	Description	Coding specification

3.3 Additional Variables of Interest (including potential confounders)

Variable Name	Description	Coding specification

4 Statistical Analysis Plan

4.1 Exploratory Data Analysis

The outcome (quarterly unemployment rate) is continuous, so we will summarize it with descriptive statistics and a histogram to assess its distribution and identify extreme values. Because our primary independent variable (the share of job postings requiring emerging AI skills) is also continuous, we will use a scatter plot of unemployment against AI adoption, with points colored and separate linear fits drawn by economy group (advanced vs. emerging/developing), to explore both the main relationship and the planned interaction. Given the skewness of both variables, we will also present this plot on log scales. Finally, we will present a Table 1 of medians and interquartile ranges for the outcome, AI adoption measures, and macroeconomic covariates by economy group, and we will summarize missingness across countries and quarters.

4.2 Statistical Model

We will fit a linear mixed-effects model with a random intercept for each economy:

$$\begin{aligned}\log(\text{Unemp}_{it}) = \beta_0 &+ \beta_1 \log(1 + \text{AI}_{it}) + \beta_2 \text{Adv}_i + \beta_3 \left[\log(1 + \text{AI}_{it}) \times \text{Adv}_i\right] \\ &+ \beta_4 \text{GDPgrowth}_{it} + \beta_5 \text{Inflation}_{it} + \beta_6 \log(\text{GDPpc}_{it}) \\ &+ \sum_y \gamma_y \text{Year}_{yt} + \sum_q \delta_q \text{Quarter}_{qt} + u_i + \varepsilon_{it}\end{aligned}$$

$$u_i \sim N(0, \sigma_u^2), \qquad \varepsilon_{it} \sim N(0, \sigma^2)$$

where:

- i indexes economies and t indexes quarters (2023 Q1 to 2025 Q4).
- Unemp_{it} is the unemployment rate (%) of economy i in quarter t.
- AI_{it} is the quarterly average share of online job postings requiring emerging AI skills (%).
- Adv_i equals 1 if economy i is classified by the IMF as an advanced economy and 0 if it is an emerging/developing economy (reference group).
- GDPgrowth_{it} is annual real GDP growth (%), Inflation_{it} is annual average CPI inflation (%), and GDPpc_{it} is GDP per capita in current U.S. dollars; these annual values are assigned to all quarters of the same year.
- Year_{yt} and Quarter_{qt} are indicator variables for calendar year (reference: 2023) and quarter (reference: Q1).
- β_0 is the intercept; β_1 is the association between AI adoption and log unemployment among emerging/developing economies; β_2 is the difference in log unemployment between advanced and emerging/developing economies when $\log(1 + \text{AI}) = 0$; β_3 is the difference in the AI slope between advanced and emerging/developing economies, so the AI slope among advanced economies is $\beta_1 + \beta_3$; β_4 to β_6 are associations with the macroeconomic covariates; γ_y and δ_q capture common year and seasonal effects.
- u_i is the random intercept for economy i, representing persistent differences in baseline unemployment across economies, and ε_{it} is the residual error.

Because the outcome is log-transformed, $100 \times (e^{\beta} - 1)$ gives the approximate percent change in the unemployment rate associated with a one-unit increase in a predictor, holding other variables constant.

4.2 Rationale for the statistical model

The outcome is continuous, and our goal is inference about the adjusted association between AI adoption and unemployment, so a linear regression framework is appropriate. We log-transform the outcome and the AI share because both are strongly right-skewed in the EDA; we use $\log(1 + \text{AI})$ because many economy-quarters have an AI share of zero.

Our data have a repeated-measures structure: each economy contributes up to 12 quarterly observations, and observations from the same economy are likely to be correlated (e.g., South

Africa has persistently high unemployment). Ordinary least squares would treat these observations as independent and understate standard errors. A random intercept for each economy accounts for this within-economy correlation. We prefer random intercepts over economy fixed effects because economy group is constant within each economy, so fixed effects would absorb it and prevent us from estimating the interaction of interest. The interaction term tests whether the association between AI adoption and unemployment differs between advanced and emerging/developing economies, and we include macroeconomic covariates because Table 1 shows that the two groups differ systematically in GDP growth, inflation, and income, which may also affect unemployment. If mixed-effects models are not appropriate for this course, we will instead fit an ordinary linear regression with cluster-robust standard errors by economy.

4.3 Model Assumptions

1. **Linearity:** log unemployment is linearly related to each continuous predictor (on the scales used in the model).
2. **Independence:** after accounting for the economy-level random intercept, residuals are independent across observations. Residuals within the same economy may still be serially correlated across consecutive quarters, which we will check.
3. **Normality:** residuals ε_{it} and random intercepts u_i are approximately normally distributed.
4. **Equal variance:** residuals have constant variance across fitted values and across economy groups.
5. **Exogeneity of random effects:** random intercepts are uncorrelated with the predictors.
6. **No severe multicollinearity** among predictors.

4.4 Model Assessment

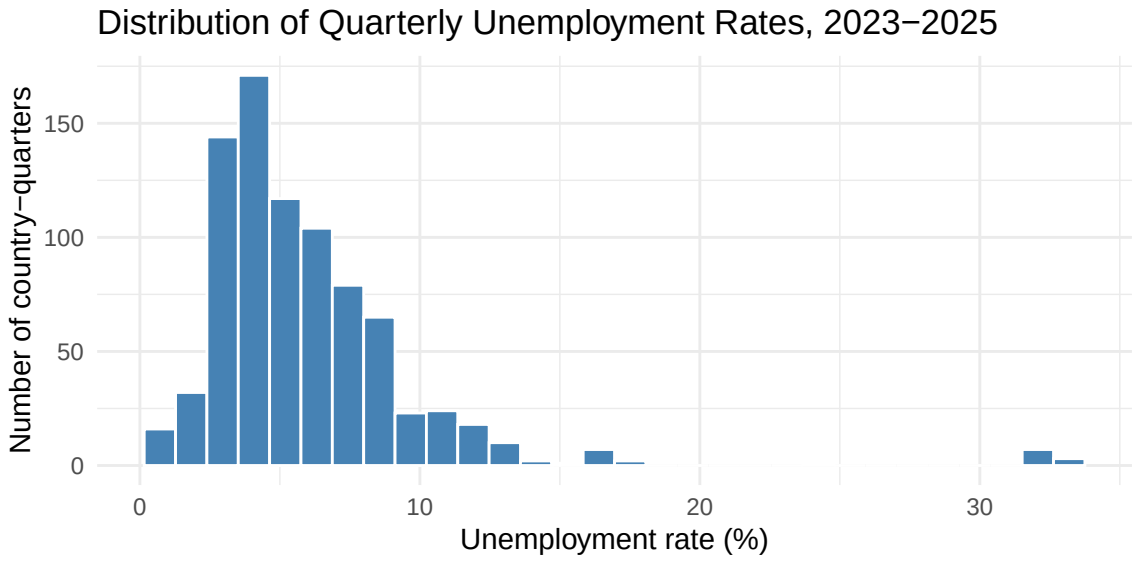
- **Linearity and equal variance:** plot residuals against fitted values and against each continuous predictor, separately by economy group.
- **Normality:** Q-Q plots of the residuals and of the estimated random intercepts.
- **Independence:** examine the autocorrelation of residuals within economies across consecutive quarters.
- **Multicollinearity:** variance inflation factors (VIFs) for the fixed effects.
- **Influential observations:** identify influential economies by refitting the model leaving out one economy at a time and by examining influence diagnostics; we will pay particular attention to the small economies with very high AI shares and to extreme inflation values (e.g., Argentina), and will consider transforming or capping inflation if needed.
- **Interaction and model fit:** compare the models with and without the interaction term using a likelihood ratio test (fit by maximum likelihood), and report the intraclass correlation and marginal and conditional R^2.
- **Sensitivity analyses:** refit the model (1) excluding economies with extreme AI shares or lower data-quality tiers, (2) using the AI share within high-exposure occupations as

the primary independent variable, and (3) using the untransformed unemployment rate, to check whether our conclusions are robust.

Exploratory Data Analysis

First, provide a snapshot of the data using the glimpse function or something similar. Then, present your EDA according to your SAP above. At minimum, you should present basic characteristics of your outcome alone (descriptive statistics and a plot) and at least one key relationship of interest, but the EDA should not exceed 3 pages. It may be useful to present a "Table 1" that gives descriptive statistics for several variables of interest. You are not required to complete data cleaning for the EDA.

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$ COUNTRY           <chr> "Brazil", "Brazil", "Brazil", "Brazil", "Brazil", "B~
$ year              <int> 2023, 2023, 2023, 2023, 2024, 2024, 2024, 2024, 2024~
$ qtr               <fct> 1, 2, 3, 4, 1, 2, 3, 4, 1, 1, 2, 3, 4, 4, 1, 3, 1, 2~
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$ ai_share_low_exp  <dbl> 0.000000000, 0.004997501, 0.025952500, 0.000000000, ~
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$ gdp_growth        <dbl> 3.242, 3.242, 3.242, 3.242, 3.419, 3.419, 3.419, 3.4~
$ inflation         <dbl> 4.594, 4.594, 4.594, 4.594, 4.367, 4.367, 4.367, 4.3~
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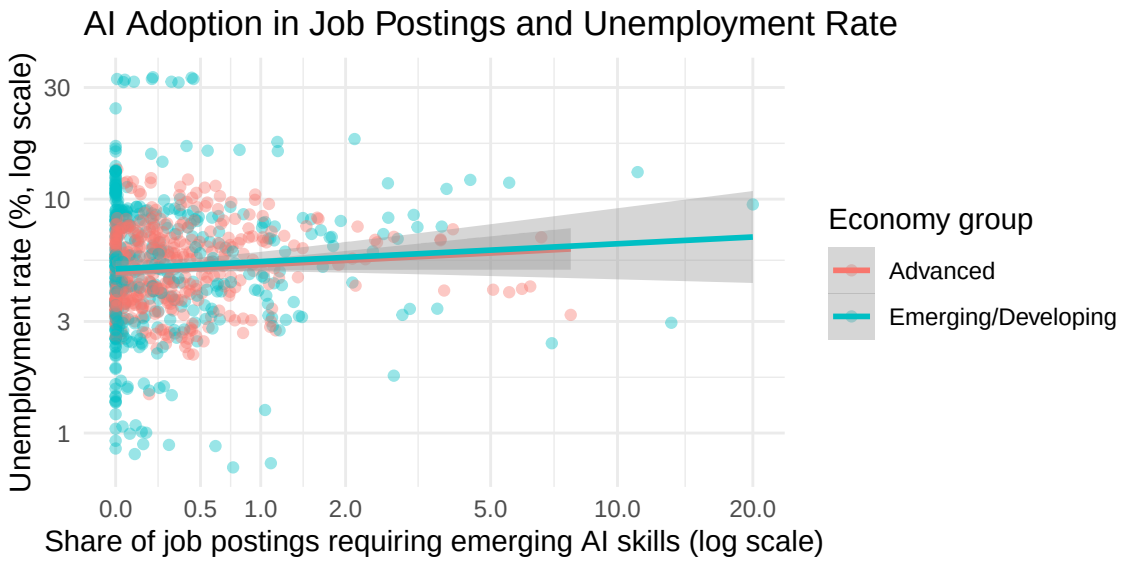


Table 1: Table 1. Median (IQR) of key variables by economy group

Characteristic	**Overall** N = 826	**Advanced** N = 403	**Emerging/Developing** N = 423
Unemployment rate (%)	5.17 (3.60, 7.39)	5.13 (3.73, 6.70)	5.26 (3.32, 8.33)
AI share, all occupations	0.17 (0.01, 0.46)	0.22 (0.05, 0.50)	0.09 (0.00, 0.41)
AI share, high-exposure occupations	0.21 (0.01, 0.59)	0.30 (0.07, 0.65)	0.12 (0.00, 0.52)
Missing	1	0	1
AI share, low-exposure occupations	0.01 (0.00, 0.15)	0.06 (0.00, 0.17)	0.00 (0.00, 0.09)
Missing	16	3	13
Real GDP growth (%)	2.13 (0.90, 3.62)	1.48 (0.81, 2.22)	3.13 (1.49, 4.08)
Inflation (%)	3.73 (2.44, 5.91)	3.02 (2.32, 4.24)	4.68 (2.87, 8.68)
GDP per capita (USD)	23,576.27 (8,700.92, 48,976.04)	49,128.50 (34,376.84, 64,848.55)	9,749.71 (4,940.26, 15,914.25)

Outcome. The analytic sample contains 826 country-quarter observations from 82 economies (403 from advanced economies and 423 from emerging/developing economies) between 2023 and 2025. The median quarterly unemployment rate is 5.17% (IQR: 3.60–7.39%). The distribution is right-skewed, with a small number of observations above 20% from South Africa and

Botswana, two economies with well-documented structural unemployment. We retain these observations and will consider a log transformation of the outcome.

Key relationship. The share of job postings requiring emerging AI skills is also highly right-skewed: most observations are below 2%, while a few smaller economies (e.g., Cyprus, Slovenia, Latvia, Bolivia, Bosnia and Herzegovina, and North Macedonia) exceed 5%. Because these economies have relatively small online job-posting markets, their high values may partly reflect measurement noise. We will assess the sensitivity of our results to these observations.

Covariates. Advanced economies have more than twice the median AI share of emerging/developing economies (0.22% vs. 0.09%), but they also differ systematically in GDP growth, inflation, and GDP per capita (Table 1). This highlights the need to adjust for these macroeconomic variables as potential confounders.

Dataset Option 2

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3.2 Primary Independent Variable(s)

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3.3 Additional Variables of Interest (including potential confounders)

Variable Name	Description	Coding specification

4 Statistical Analysis Plan

4.1 Exploratory Data Analysis

Detail how you will investigate the data, including basic characteristics of the outcome and independent variable(s) of interest (e.g., "The outcome is continuous, so we will generate a histogram of the outcome. Then, we will generate a stacked histogram/scatter plot to explore the relationship between the outcome and the independent variable")

4.2 Statistical Model (write in mathematical notation and define each term)

4.2 Rationale for the statistical model (why are you choosing to use this type of model? What about your data structure/characteristics are you accommodating with this model type?)

4.3 Model Assumptions

4.4 Model Assessment (how will you assess the assumptions and model performance?)

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Team Charter

1. When will you meet as a team to work on the project components? Will these meetings be held in person or virtually?
2. What is your group policy on missing team meetings (e.g., how much advance notice should be provided)?
3. How will your team communicate (email, Slack, text messages)? What is your policy on appropriate response time (within a certain number of hours? Nights/weekends?)?